

ME 504

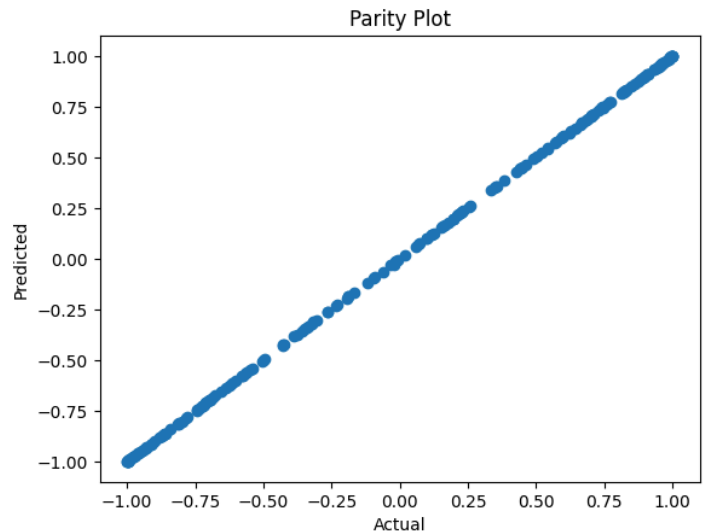
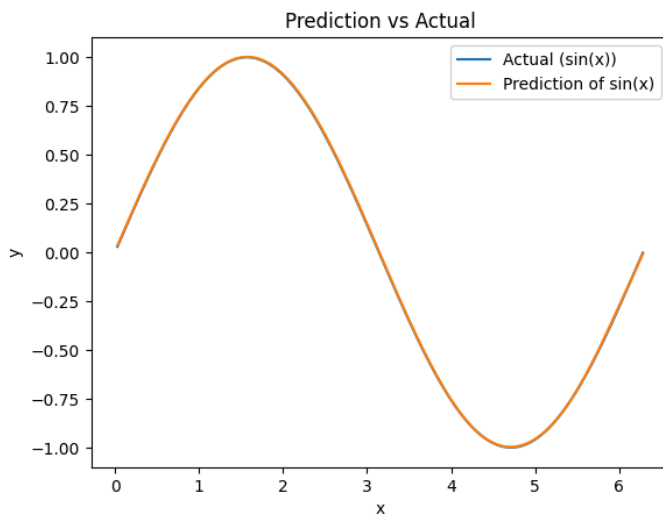
Assignment 4 - Result Report

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2024MEB1355

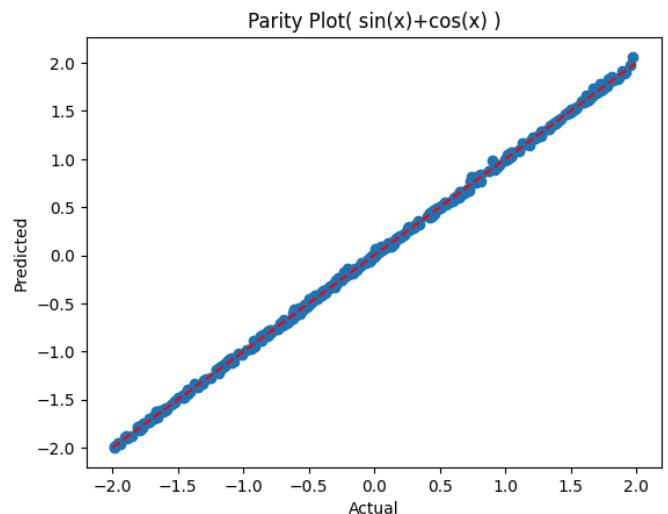
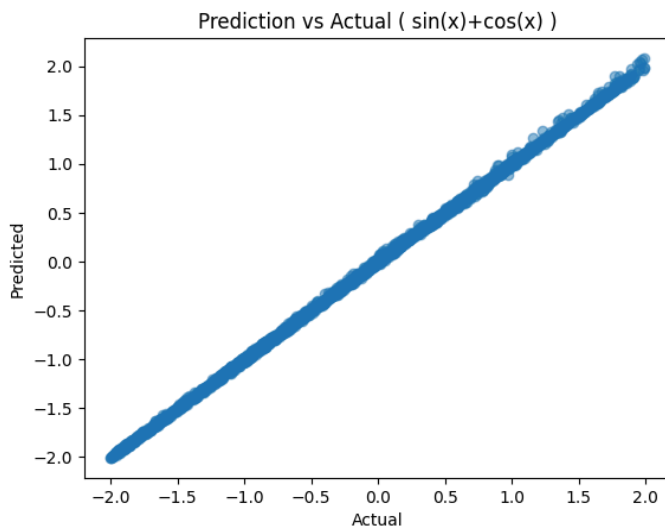
Question 1

1. Verify your regression program by plotting the predicted values against the analytical solution for both the functions. Also verify using the parity plot.

Function 1 = $\sin x$

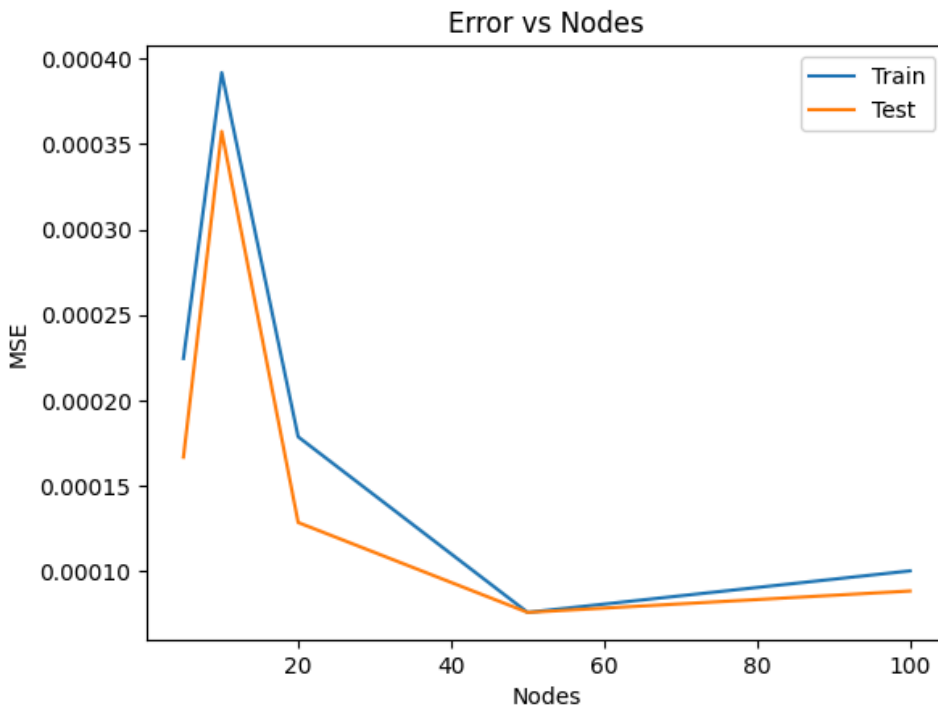


Function 2 = $\sin x + \cos x$



In both the above cases, the deviation from $y = x$ is minimal in the parity plot, signifying minimum amount of error.

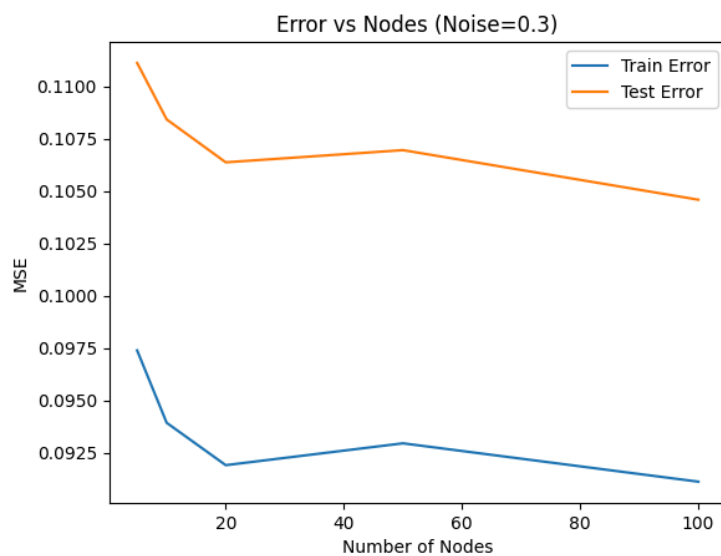
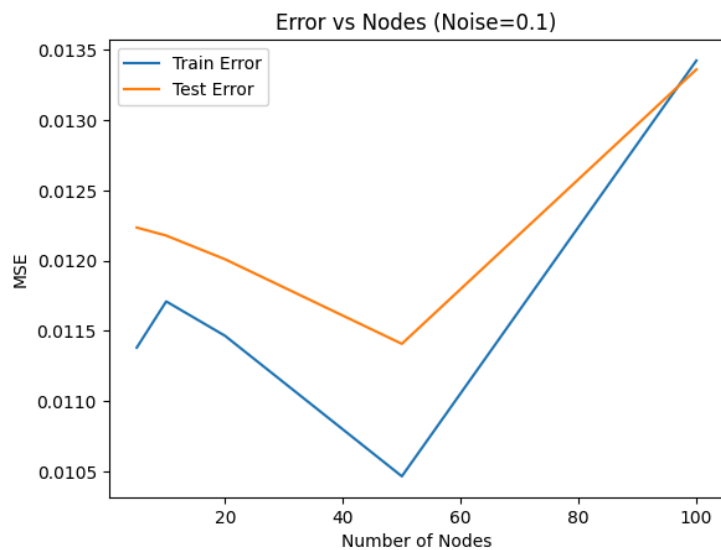
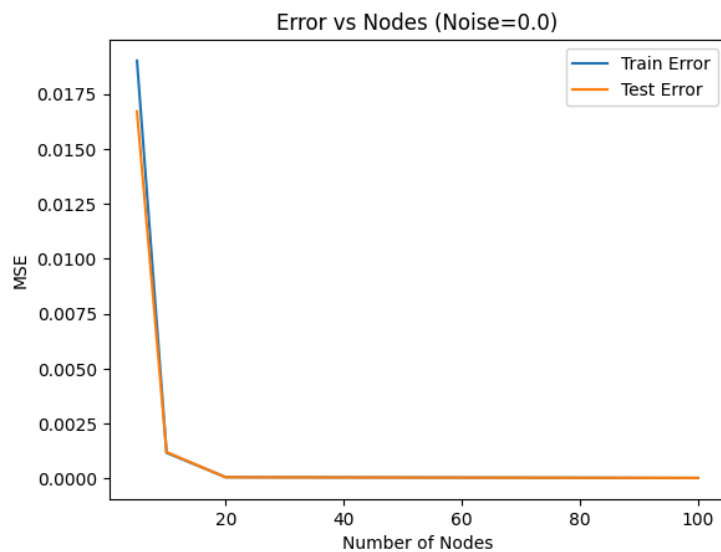
- 2. Plot the training and testing error with respect to the number of nodes in layers. From this plot identify the optimal number of nodes. You can take any number of hidden layer for this case.**



The optimum number of nodes as seen in the graph is 50.

- Small nodes go into underfitting region
- Large nodes go into overfitting region

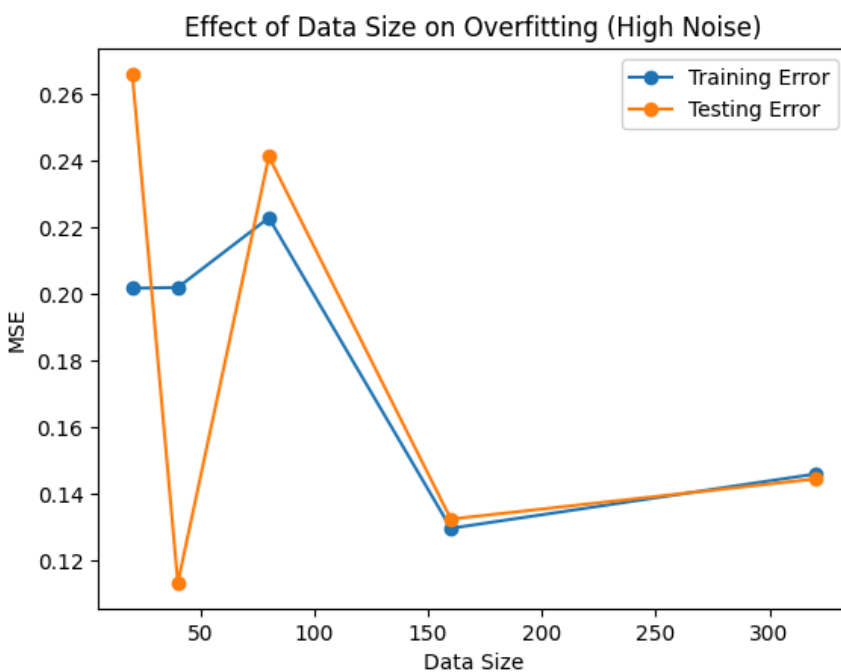
**3. Carry out the point no. 2 exercise for three different levels of noise.
Does the noise affect the conclusions of point-2.**



Increasing noise in the dataset leads to higher training and testing errors. The presence of noise makes it difficult for the model to learn the underlying function accurately.

Additionally, the optimal number of nodes shifts.

4. **Take the case of the highest noise, considered in point-3. Further, take the NN that lies in the over fitting region. Now increase the data size in the geometric progression of two, and plot the training as well as testing error with respect to the data size.**



With small data sizes, the model shows significant overfitting, characterized by low training error and high testing error.

As the dataset size increases, the gap between training and testing error reduces, indicating improved generalization.

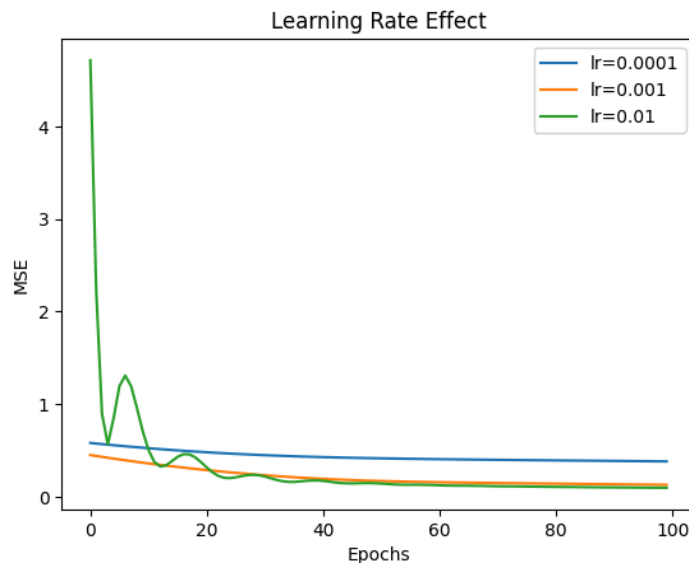
5. Take the point similar to the point-4, so that essentially you are in the over-fitting regime. Now use the L2 regularization to control the capacity of the model.

```
WITHOUT regularization:  
Train MSE: 0.0010160936508327723  
Test  MSE: 0.06599107384681702  
  
WITH regularization:  
Train MSE: 0.009282729588449001  
Test  MSE: 0.0777786448597908
```

L2 regularization penalizes large weights and reduces model complexity. Training error increases whereas testing error should decrease in the overfitting region. Here, since it is borderline overfitting, the testing error increases but in very less quantity as compared to the training error.

It was observed that too high of a regularisation parameter leads to underfitting due to over-regularisation. So we use $\lambda = 0.00001$

6. Play with the learning rate and show how it affects the overall performance.



A very high learning rate leads to unstable training, while a very low learning rate results in slow convergence. An intermediate learning rate provides the best performance.

7. Play with the number of layers, and see its effect on the training and testing error.

```
Layers=1, Test Loss=0.1482  
Layers=2, Test Loss=0.0624  
Layers=3, Test Loss=0.0639  
Layers=4, Test Loss=0.0615
```

Increasing the number of layers improves the model's ability to capture complex patterns.

8. Take different choices of activation functions, and analyze its effect.

```
relu → Test Loss = 0.1167  
tanh → Test Loss = 0.0616  
sigmoid → Test Loss = 0.2215
```

Among the tested activation functions, tanh performs best for approximating the sine function due to its similar range and pattern. ReLU and sigmoid show comparatively poorer performance, with sigmoid being the worst choice.

Question 2

We calculate the